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980284 8/06/2021

1)

x_1 tonnellate di A
x_2 tonnellate di B

$$\max z = x_1 + 2x_2$$
$$2x_1 + 3x_2 \leq 16$$
$$x_1 \geq 1$$
$$x_2 \geq 2$$
$$x_1, x_2 \geq 0$$

FORMA STANDARD

$$\min z = -x_1 - 2x_2$$
$$+2x_1 + 3x_2 + x_3 = 16$$
$$x_1 - x_4 = 1$$
$$x_2 - x_5 = 2$$
$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

2)

Fase 1

$$\begin{array}{cccccccc} & & & & & & x_{a1} & x_{a2} \end{array}$$
$$\left[\begin{array}{c|ccccc|cc} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ \hline 16 & 2 & 3 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & -1 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{c|ccccc|cc} -3 & -1 & -1 & 0 & 1 & 1 & 0 & 0 \\ \hline 16 & 2 & 3 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & -1 & 0 & 1 \end{array}\right]$$

$B = \{A_3, A_6, A_2\}$
$x = (0,0,16,0,0,1,2)$
$\alpha = (0,0)$

$$\left[\begin{array}{c|ccccc|cc} -2 & 0 & -1 & 0 & 0 & 1 & 1 & 0 \\ \hline 14 & 0 & 3 & 1 & 2 & 0 & -2 & 0 \\ 1 & 1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & -1 & 0 & 1 \end{array}\right]$$

$x = (1,0,14,0,0,0,2)$
$\beta = (1,0)$

$$\left[\begin{array}{c|ccccc|cc} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ \hline 8 & 0 & 0 & 1 & 2 & 3 & -2 & -3 \\ 1 & 1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & -1 & 0 & 1 \end{array}\right]$$

$x = (1,2,8,0,0,0,0)$
$\gamma = (1,2)$

Fase 2

$$\left[\begin{array}{c|ccccc|cc} 0 & -1 & -2 & 0 & 0 & 0 & 0 & 0 \\ \hline 8 & 0 & 0 & 1 & 2 & 3 & -2 & -3 \\ 1 & 1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & -1 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{c|ccccc|cc} 5 & 0 & 0 & 0 & -1 & -2 & 1 & 2 \\ \hline 8 & 0 & 0 & 1 & 2 & 3 & -2 & -3 \\ 1 & 1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & -1 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{c|ccccc|cc} 31/3 & 0 & 0 & 2/3 & 1/3 & 0 & -1/3 & 0 \\ \hline 8/3 & 0 & 0 & 1/3 & 2/3 & 1 & -2/3 & -1 \\ 1 & 1 & 0 & 0 & -1 & 0 & 1 & 0 \\ 14/3 & 0 & 1 & 1/3 & 2/3 & 0 & -2/3 & 0 \end{array}\right]$$

$x = \left(1, \frac{14}{3}, 0, 0, \frac{8}{3}\right)$

$\delta = \left(1, \frac{14}{3}\right)$

3) Si producono 1 tonnellate A e $\frac{14}{3}$ tonnellate di B
Il profitto è $\frac{31}{3}$ volte quello di 1 tonnellata di A

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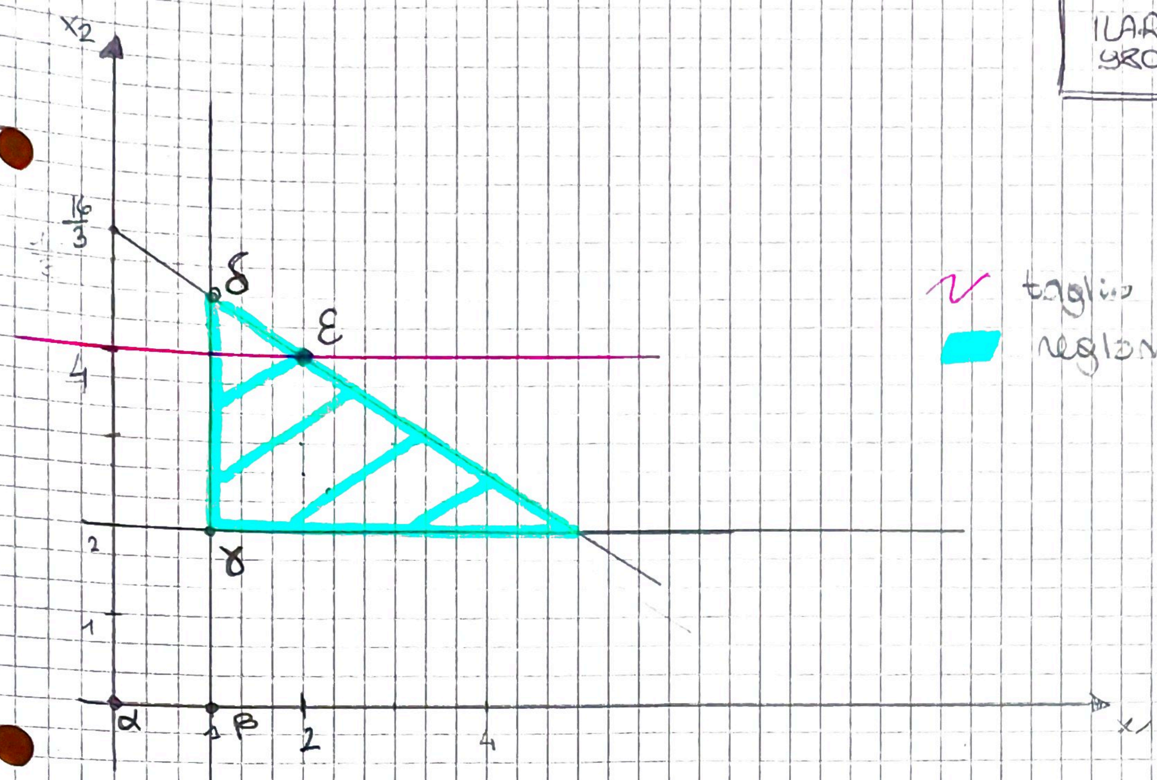
4)



taglio

regione ammissibile

5)

$$
\begin{aligned}
\max w = \; & 16\pi_1 + \pi_2 + 2\pi_3 \\
& 2\pi_1 + \pi_2 \le -1 \\
& 3\pi_1 + \pi_3 \le -2 \\
& \pi_1 \le 0 \\
& -\pi_2 \le 0 \\
& -\pi_3 \le 0 \\
& \pi_1,\ \pi_2,\ \pi_3 \gtrless 0
\end{aligned}
$$

Base iniziale

$$
\begin{array}{ccc} A_3 & A_6 & A_7 \end{array}
\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

Tableau finale

$$
\begin{array}{ccc} A_3 & A_6 & A_7 \end{array}
\begin{bmatrix} 2/3 & -1/3 & 0 \\ \vdots & \vdots & \vdots \end{bmatrix}
$$

$A_3 \quad \pi_1 = 0 - 2/3 = -\frac{2}{3}$

$A_6 \quad \pi_2 = 0 - (-1/3) = +\frac{1}{3}$

$A_7 \quad \pi_3 = 0 - 0 = 0$

Soluzione duale $\pi = \left(-\frac{2}{3}, \frac{1}{3}, 0\right)$

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6) Branch and bound $\quad UB_0 = \left\lfloor \frac{31}{3} \right\rfloor = 10$

$x_2 \leq \left\lfloor \frac{14}{3} \right\rfloor$

$x_2 \leq 4$

$x_2 + x_6 = 4$

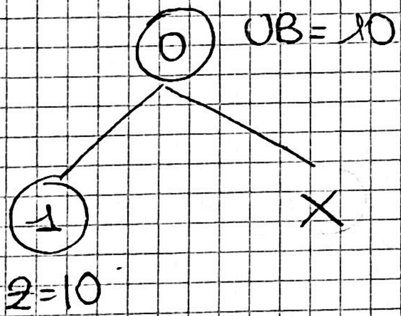
$$\left[\begin{array}{c|cccccc}
31/3 & 0 & 0 & 2/3 & 1/3 & 0 & 0 \\
\hline
8/3 & 0 & 0 & 1/3 & 2/3 & 1 & 0 \\
1 & 1 & 0 & 0 & -1 & 0 & 0 \\
14/3 & 0 & 1 & 1/3 & 2/3 & 0 & 0 \\
-2/3 & 0 & 0 & -1/3 & -2/3 & 0 & 1
\end{array}\right]$$

$$\left[\begin{array}{c|cccccc}
10 & 0 & 0 & 1/2 & 0 & 0 & 1/2 \\
\hline
2 & 0 & 0 & 0 & 0 & 1 & 1 \\
2 & 1 & 0 & 1/2 & 0 & 0 & -3/2 \\
4 & 0 & 1 & 0 & 0 & 0 & 1 \\
1 & 0 & 0 & 1/2 & 1 & 0 & -3/2
\end{array}\right]$$

$x = (2, 4, 0, 1, 2)$

$z = 10$

$\bar{x} = (2, 4)$



taglio perché ho già la soluzione ottima

Soluzione intera:

si producono 2 tonnellate di A, 4 tonnellate di B con profitto pari a 10 volte quello di 1 tonnellata di A